

## ① REQUIREMENTS — canonical enterprise-Q&A RAG workload

~2,000 users · ~10/40 rps · 9.2K-in/300-out · quality ~70B



## ② CONSTRAINTS — hard gates

memory (residency  $\leq$  host) · latency (TTFT/TPOT) · economics (per-host ceiling)



## ③ GENERATE CANDIDATES

(a) 8×H100 + prefix cache · (b) P/D-disaggregated 2-pool · (c) KV-quantized 7B



## ④ TEST AGAINST memory · latency · economics

### (a) 8×H100 + prefix cache

MEM ✓ · latency ✓ · cost ✓ | but prefill ~19.8K  $\ll$  ~92K needed

✗ capacity (needs ~5 hosts)

### (b) P/D-disaggregated 2-pool

MEM ✓ · latency ✓ · 2 hosts + fabric (~2× capex)

✓ survives as the at-scale answer

### (c) KV-quantized 7B

MEM ✓ · cheapest (1 H100) | but fails the QUALITY gate

✗ EXCLUDED



## ⑤ OUTCOME — SURVIVORS

(a) per-request-latency baseline  
(b) at-scale P/D answer — carried to Ch.13-14

## ✗ REJECTED

(c) fails quality gate  
→ excluded from evaluation